

TOPIC NAME : _____

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MOHSIN IBNA HOSSAIN
AIUB, MATH 6

FINAL
Term

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Time Series Analysis
on
Choo8

Time Series:-

⇒ It is a Sequence of data points recorded in time order, often taken at successive equally spaced points in time.

Components:-

1. Trend:- General Upward or downward direction may not be consistent over the time.
2. Seasonality:- Predictable pattern recurring at regular intervals, usually within a year.
3. Cycles:- periodic up and down not tied to seasons, includes business cycle phase.
4. Irregular Fluctuation:- Unpredictable variations due to sudden events like wars, Natural disasters or strikes.

Forecasting Methods

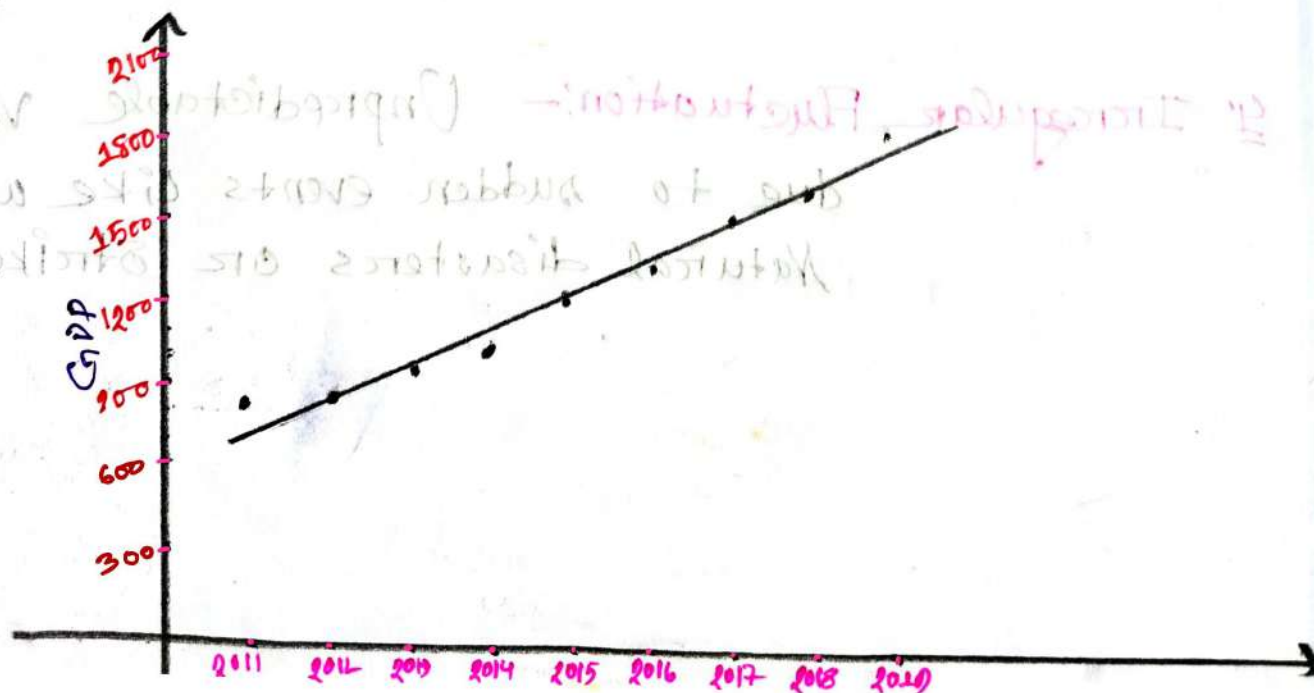
1. Graphical Method.
2. Semi-Average Method.
3. Moving-Average Method.
4. Single Exponential Method.
5. Least Squares Method.

Graphical Method:—

Ex. Example: 8.1:—

⇒ GDP per capital in DD is listed below:—

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GDP	836	857	952	1085	1211	1350	1517	1625	1827



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Semi-Average Method!—

Example: 8.2

Year	2008	2009	2010	2011	2012	2013
Income	46.17	51.65	63.81	70.00	84.91	91.64

⇒ Calculate the trend values using semi-average methods for the income from the forest department.

⇒ Solution!—

Year	Income	3 Year Semi Avg	Trend Values
2008	46.17	$= \frac{46.17 + 51.65 + 63.81}{3}$ $= 53.877$	
2009	51.65		
2010	63.81		
2011	70.00	$= \frac{70.00 + 84.91 + 91.64}{3}$ $= 82.513$	
2012	84.91		
2013	91.64		

⇒ Difference of central years = $2012 - 2009 = 3$

⇒ Difference of Semi Avg Values = $82.513 - 53.877$
 $= 28.636$

⇒ Increase in trend values for one year = $28.636 / (3/1)$
 $= 9.545$

Semi-Average Method!—

Example: 8.2

Year	2008	2009	2010	2011	2012	2013
Income	46.17	51.65	63.81	70.00	84.01	91.64

⇒ Calculate the trend values using semi-average methods for the income from the Forest department.

⇒ Solution!—

Year	Income	3 Year Semi Avg	Trend Values
2008	46.17	$= \frac{46.17 + 51.65 + 63.81}{3}$ $= 53.877$	$53.877 - 0.545 = 44.002$
2009	51.65		$44.002 + 0.545 = 53.877$
2010	63.81		$53.877 + 0.545 = 63.422$
2011	70.00	$= \frac{70.00 + 84.01 + 91.64}{3}$ $= 82.513$	$63.422 + 0.545 = 72.067$
2012	84.01		$72.067 + 0.545 = 82.512$
2013	91.64		$82.512 + 0.545 = 92.057$

⇒ Difference of central years = $2012 - 2009 = 3$

⇒ Difference of Semi Avg values = $82.513 - 53.877$
 $= 28.636$

⇒ Increase in trend values for one year = $28.636 / (3/1)$
 $= 0.545$

① Now Calculate the trend value for 2014.

⇒ ∴ Trend value for 2014

$$= 92.057 + 0.545$$

$$= 92.602$$

② Also calculate the trend value for 2007.

⇒ ∴ Trend Value for 2007:-

$$= 44.332 - 0.545$$

$$= 34.787$$

Ans:-

Example: 8.3:-

Year	1951	1961	1971	1981	1991	2001	2011
Income	302.2	336.0	412	484.1	558.6	624.1	721.4

⇒ ① Find the trend values using Semi Average method.

② Also Calculate the trend value for 2021.

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⇒

②

Central
Year

Central
Year.

Year	Income	3 Year-Semi Avg	Trend values
1951	301.2	$= \frac{301.2 + 336.0 + 412}{3}$ $= 350.033$	
1961	336.0		
1971	412		
1981	384.1		
1991	558.6	$= \frac{558.6 + 624.1 + 721.4}{3}$ $= 634.7$	
2001	624.1		
2011	721.4		

⇒ Difference of central years = $2001 - 1961 = 40$

⇒ Difference of 3 Year Semi Avg = $634.7 - 350.033$
 $= 284.667$

∴ Increase in trend value over 10 Year = $284.667 / (40/10)$
 $= 71.167$

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①

Year	Income	3 Year Semi Avg	Trend values
1951	301.2	$= \frac{301.2 + 336.0 + 412}{3}$ $= 350.033$	$350.03 - 71.17 = 278.86$
1961	336.0		$278.86 + 71.17 = 350.03$
1971	412		$350.03 + 71.17 = 421.2$
1981	384.1		$421.2 + 71.17 = 492.37$
1991	558.6	$= \frac{558.6 + 624.1 + 721.4}{3}$ $= 634.7$	$492.37 + 71.17 = 563.54$
2001	624.1		$563.54 + 71.17 = 634.71$
2011	721.4		$634.71 + 71.17 = 705.88$

⇒ Difference of central years = $2001 - 1961 = 40$

⇒ Difference of 3 Year Semi Avg = $634.7 - 350.033$
 $= 284.667$

∴ Increase in trend value over 10 Year = $284.667 / (40/10)$
 $= 71.167$

②

⇒ ∴ Trend value for 2021 =

$$= 705.88 + 71.17$$

$$= 777.05$$

Ans:

Exercise 8

8.1 Show the principle of Moving average for the given data (Using 3 years).

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Revenue	5.0	8.0	7.0	8.0	8.0	9.0	7.0	9.0	5.0	7.0	5.0	8.0

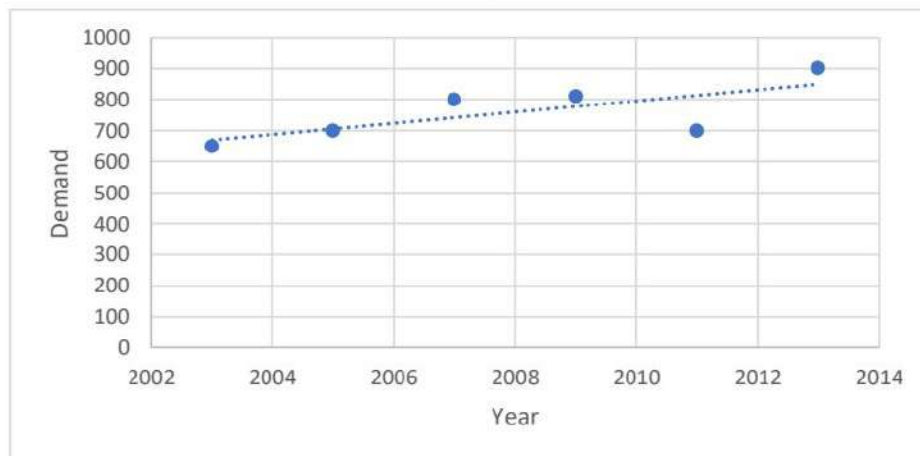
→ **Solution:**

Month	Revenue	Moving Avg for Three Months
Jan	5.0	X
Feb	8.0	$(5 + 8 + 7) / 3 = 6.67$
Mar	7.0	$(8 + 7 + 8) / 3 = 7.67$
Apr	8.0	$(7 + 8 + 8) / 3 = 7.67$
May	8.0	$(8 + 8 + 9) / 3 = 8.33$
Jun	9.0	$(8 + 9 + 7) / 3 = 8.00$
Jul	7.0	$(9 + 7 + 9) / 3 = 8.33$
Aug	9.0	$(7 + 9 + 5) / 3 = 7.00$
Sep	5.0	$(9 + 5 + 7) / 3 = 7.00$
Oct	7.0	$(5 + 7 + 5) / 3 = 5.67$
Nov	5.0	$(7 + 5 + 8) / 3 = 6.67$
Dec	8.0	X

8.2 Fit trend lines by the methods of i) graphical ii) semi-averages for the given data.

Year	2003	2005	2007	2009	2011	2013
Demand	650	700	800	810	700	900

(i)



(ii)

Year	Demand	3-year Semi Avg	Trend Values
2003	650	$\frac{650 + 700 + 800}{3} = 716.67$	
2005	700		
2007	800		
2009	810	$\frac{810 + 700 + 900}{3} = 803.33$	
2011	700		
2013	900		

Difference between the central years = 2011 – 2005 = 6

Difference between the semi-averages = 803.33 – 716.67 = 86.66

Increase in trend value for 2 years = 86.66 / (6/2) = 28.89

Year	Demand	3-year Semi Avg	Trend Values
2003	650	$\frac{650 + 700 + 800}{3} = 716.67$	$716.67 - 28.89 = 687.78$
2005	700		$687.78 + 28.89 = 716.67$
2007	800		$716.67 + 28.89 = 745.56$
2009	810	$\frac{810 + 700 + 900}{3} = 803.33$	$745.56 + 28.89 = 774.45$
2011	700		$774.45 + 28.89 = 803.33$
2013	900		$803.33 + 28.89 = 832.23$

8.3 Fit a trend line by the method of semi-averages for the given data.

Year	1990	1991	1992	1993	1994	1995	1996
Sales	15	11	20	10	15	25	35

(ii)

Year	Sales	3-year Semi Avg	Trend Values
1990	15	$\frac{15 + 11 + 20}{3} = 15.33$	
1991	11		
1992	20		
1993	10	$\frac{15 + 25 + 35}{3} = 25$	
1994	15		
1995	25		
1996	35		

Difference between the central years = 1995 – 1991 = 4

Difference between the semi-averages = 25 – 15.33 = 9.67

Increase in trend value for 1 years = 9.67 / (4/1) = 2.418

Year	Sales	3-year Semi Avg	Trend Values
1990	15	$\frac{15 + 11 + 20}{3} = 15.33$	$15.33 - 2.418 = 12.912$
1991	11		$12.912 + 2.418 = 15.33$
1992	20		$15.33 + 2.418 = 17.748$
1993	10		$17.748 + 2.418 = 20.166$
1994	15	$\frac{15 + 25 + 35}{3} = 25$	$20.166 + 2.418 = 22.548$
1995	25		$22.548 + 2.418 = 25.002$
1996	35		$25.002 + 2.418 = 27.42$

8.4 Measure the trend by the method of semi-averages by using the table given below. Estimate the value for the year 1994-1995.

Years	Value in Million
1984 – 85	18.6
1985 – 86	22.6
1986 – 87	38.1
1987 – 88	40.9
1988 – 89	41.4
1989 – 90	40.1
1990 – 91	46.6
1991 – 92	60.7
1992 – 93	57.2
1993 – 94	53.4

Stochastic
Process
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Stochastic process: $(X(t))$

☒ Discrete-time process:—

$$\Rightarrow \{X(t) : t \in T\}, \quad T = \{0, 1, 2, \dots\}$$

☒ Continuous time process:—

$$\Rightarrow \{X(t) : t \in T\}, \quad T = [0, \infty]$$

Counting process:—

☒ Stochastic process + total No:

$$\Rightarrow N(t) \geq 0$$

$$\Rightarrow N(t) = \text{integer}$$

$$\Rightarrow \text{if } s \leq t \text{ then } N(s) \leq N(t)$$

$$\Rightarrow \text{if } s < t \text{ then } N(t) - N(s) = \text{No. of event occurred in between } [s, t]$$

Markov Process:—

$$\Rightarrow \text{Stochastic process + Discrete-time process + } P_{ij} \geq 0 \text{ \& } \sum P_{ij} = 1$$

Property

1: $X_t \rightarrow X_{t+1}$ (Future depends on present)

2: Transition Matrix $\rightarrow$ Square matrix & $\sum P_{ij} = 1$

Poisson process:-

$$1. P(T > t) = e^{-\lambda t}$$

$$2. P(T < t) = 1 - e^{-\lambda t}$$

$$3. P(t_1 < T < t_2) = e^{-\lambda t_1} - e^{-\lambda t_2}$$

$$4. E(S_n) = \frac{n}{\lambda}$$

Example:-

⇒ Two state Markov process is given as:-

State: 0: Today Sunny Tomorrow Sunny → $p = 0.6$

State: 1: Today Not Sunny Tomorrow Sunny → $p = 0.5$

Q Find the probability (p) that the sunny day continue for four days, starting from today.

Q

In given:-

⇒ The transition p. matrix:

$$P = \begin{matrix} & \begin{matrix} S & NS \end{matrix} \\ \begin{matrix} S \\ NS \end{matrix} & \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix} \end{matrix}$$

We need P_{00}^4 in P^4

$$\therefore p^2 = p^1 \cdot p^1$$

$$= \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix}$$

$$\therefore p^4 = p^2 \cdot p^2$$

$$= \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix} \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix}$$

$$= \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix}$$

$\therefore$ The required probability = 56%

Example —

$\Rightarrow$ Communication system sends two signal 0 & 1.

0 reaches to its destination properly with probability 0.4.

(a) Find 0 sent from the station will reach destination at 5th stage.

(a)

In given:-

$\Rightarrow$ the transition p. matrix:-

$$p^1 = \begin{matrix} & \begin{matrix} R & N \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{bmatrix} \end{matrix}$$

$\Rightarrow$ To reach the destination at 5th step, it needs further $(5-1) = 4$ steps.

$$\therefore P^2 = \begin{bmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{bmatrix} \begin{bmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix}$$

$$\therefore P^4 = \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix} \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix}$$

$$= \begin{matrix} \text{R} & \text{N} \\ 0 & 1 \end{matrix} \begin{bmatrix} 0.50 & 0.40 \\ 0.40 & 0.50 \end{bmatrix}$$

$\therefore$ The required probability = 50%

Example:

$\Rightarrow$ Signal sent with poisson rate 5 per hour. Find probability of elapsed time between the Entrance of 5th & 6th, signal is.

① $P(\text{less 1 hour})$

② $P(\text{more than half hour})$

③ $P(\text{in between 30 min to 60 min})$

①

$\Rightarrow$ I.A. given:—

$$\lambda = 5$$

$$\therefore P(T < 1 \text{ hr}) = 1 - e^{-5 \cdot 1} = 0.99 = 99\%$$

(b)We know:-

$$P(T > t) = e^{-\lambda t}$$

$$\therefore P(T > 0.5 \text{ hr}) = e^{-5 \times 0.5}$$

$$= 0.082$$

$$= 8.2\%$$

(c)We know:-

$$P(t_1 < T < t_2) = e^{-\lambda t_1} - e^{-\lambda t_2}$$

$$\therefore P(0.5 < T < 1 \text{ hr}) = e^{-5 \times 0.5} - e^{-5 \times 1}$$

$$= 0.075$$

$$= 7.5\%$$

(d) Find the expected time until the n th signal reach its goal.

(d)In given:-

$$n = 0, \lambda = 5$$

We know:-

$$E(s_n) = \frac{n}{\lambda} = \frac{0}{5} = 0 \text{ hr.}$$

Example:-

⇒ Emails enter & sent from a cyber cafe reaches to the next stage with probability 0.7.

① Find the probability that an email sent from cyber cafe reach properly at 4th stage.

①

⇒ We know:-

The transition probability Matrix:-

$$P^1 = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

In given:-
mo
 $p = 0.7$

$$\therefore P^1 = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

It will reach to the destination at $(4-1) = 3$ step.

$$\begin{aligned} \therefore P^2 &= P^1 \cdot P^1 = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \\ &= \begin{bmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \therefore P^3 &= P^2 \cdot P^1 = \begin{bmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \\ &= \begin{bmatrix} 0.53 & 0.47 \\ 0.47 & 0.53 \end{bmatrix} \end{aligned}$$

∴ The required probability = 0.53
= 53%

Exercise 7

- 7.4. Customers enter to a mobile repairing shop from 9 – 10 and from 10 – 11 with probability 0.6. Customers also enter to the shop from 10 – 11 but not from 9 – 10 with probability 0.8. Find the probability that the customers will enter to the repairing shop up to 2 PM starting from 9 AM.

⇒ We know:—

The transition probability Matrix:—

$$P^1 = \begin{matrix} & \begin{matrix} B \\ B' \end{matrix} \\ \begin{matrix} A \\ A' \end{matrix} & \begin{bmatrix} 0.6 & 0.4 \\ 0.8 & 0.2 \end{bmatrix} \end{matrix}$$

We need P_{00}^5 in P^5

AB = Enter 9-10 from 10-11
 AB' = Not Enter 9-10 but Enter 10-11
 $A'B$ = Enter 9-10 but Not E. 10-11
 $A'B'$ = Not E. 9-10 & Not E. 10-11

$$\therefore P^2 = P^1 \cdot P^1 = \begin{bmatrix} 0.6 & 0.4 \\ 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.8 & 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.68 & 0.32 \\ 0.64 & 0.36 \end{bmatrix}$$

$$\therefore P^3 = P^2 \cdot P^1 = \begin{bmatrix} 0.68 & 0.32 \\ 0.64 & 0.36 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.8 & 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.664 & 0.336 \\ 0.672 & 0.328 \end{bmatrix}$$

$$\therefore P^5 = P^3 \cdot P^2 = \begin{bmatrix} 0.664 & 0.336 \\ 0.672 & 0.328 \end{bmatrix} \begin{bmatrix} 0.68 & 0.32 \\ 0.64 & 0.36 \end{bmatrix}$$

$$= \begin{bmatrix} 0.67 & 0.33 \\ 0.67 & 0.33 \end{bmatrix}$$

$\therefore$ The required probability = 0.67
 = 67%

- 7.5. Customers enter to a mobile repairing shop from 9 – 10 and from 10 – 11 with probability 0.6. Customers also enter to the shop from 10 – 11 but not from 9 – 10 with probability 0.4. Find the probability that the customers will enter to the repairing shop up to 1 PM starting from 9 AM.

⇒ We know —

The transition probability matrix —

$$P^+ = \begin{matrix} & \begin{matrix} B \\ 0.6 \end{matrix} & \begin{matrix} B' \\ 0.4 \end{matrix} \\ \begin{matrix} A \\ 0.4 \end{matrix} & \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix} & \begin{matrix} A' \\ 0.6 \end{matrix} \end{matrix}$$

We need P_{00}^4 in P^4

$$\therefore P^2 = P^+ \cdot P^+ = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix}$$

$$\therefore P^4 = P^2 \cdot P^2 = \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix} \begin{bmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{bmatrix}$$

$$= \begin{bmatrix} 0.50 & 0.49 \\ 0.49 & 0.50 \end{bmatrix}$$

∴ The required probability = 0.50
= 50%

- 7.6. E. mails enter to a cybercafé at a Poisson rate $\lambda = 2$ per minute. Find the probability that the elapsed time between the entrance of 10th and 11th mail is (i) more than 1 minute, (ii) less than 2 minutes, (iii) between 1 to 2 minutes.

$\Rightarrow$ Is given:—

$$\lambda = 2$$

①

We know:—

$$\begin{aligned} P(T > 1) &= e^{-\lambda t} \\ &= e^{-2 \times 1} \\ &= 0.135 = 13.53\% \end{aligned}$$

②

We know:—

$$\begin{aligned} P(T < 2) &= 1 - e^{-\lambda t} \\ &= 1 - e^{-2 \times 2} \\ &= 0.981 = 98.17\% \end{aligned}$$

③

We know:—

$$\begin{aligned} P(1 < T < 2) &= e^{-\lambda \times 1} - e^{-\lambda \times 2} \\ &= e^{-2 \times 1} - e^{-2 \times 2} \\ &= 0.117 \\ &= 11.7\% \end{aligned}$$

- 7.7. The customer enters to a bank at Poisson rate $\lambda = 25$ per hour. Find the probability that the elapsed time between the entrance of 10th and 11th customer is (i) less than 2 minutes, (ii) more than 5 minutes, (iii) 3 to 5 minutes.

⇒ Is given

$\lambda = 25$ per hour

(i)

we know:-

$$P(T < t) = 1 - e^{-\lambda t}$$

$$\begin{aligned} \therefore P(T < 2 \text{ min}) &= 1 - e^{-25 \times \frac{2}{60}} \\ &= 0.56 \\ &= 56\% \end{aligned}$$

(ii)

we know:-

$$\begin{aligned} P(T > 5 \text{ min}) &= e^{-\lambda t} \\ &= e^{-25 \times \frac{5}{60}} \\ &= 0.12 = 12\% \end{aligned}$$

(iii)

we know:-

$$\begin{aligned} P(3 \text{ min} < T < 5 \text{ min}) &= e^{-\lambda t_1} - e^{-\lambda t_2} \\ &= e^{-25 \times \frac{3}{60}} - e^{-25 \times \frac{5}{60}} \\ &= 0.161 \\ &= 16.1\% \end{aligned}$$

- 7.8. E-mails enter to a cybercafé at a Poisson rate $\lambda = 2$ per minute. Find the expected time until the entrance of 10th mail.

$\Rightarrow$ In given:-

$$\lambda = 2 \text{ per min}$$

$$n = 10$$

we know:-

$$E(S_n) = \frac{n}{\lambda} = \frac{10}{2} = 5 \text{ min}$$

Ans:-

- 7.9. The customer enters to a bank at Poisson rate $\lambda = 25$ per hour. Find the expected time until the entrance of 50th customer.

$\Rightarrow$ In given:-

$$\lambda = 25 \text{ per hr}$$

$$n = 50$$

we know:-

$$E(S_n) = \frac{n}{\lambda} = \frac{50 \times 25}{25} = 2 \text{ hour}$$

Ans:-

7.10 Consider a communication system which transmits the digits 0 and 1 through several stages. At each stage the probability that the same digit will be received by the next stage, as transmitted, is 0.75. What is the probability that a 0 that is entered at the first stage is received as a 0 by the 5th stage?

⇒ We know:—

The transition probability matrix:—

$$P^1 = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix} \end{matrix}$$

We need P_{00}^4 in P^4 Since, 5th stage

$$\therefore P^2 = P^1 \cdot P^1 = \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix} \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix}$$

$$= \begin{bmatrix} 0.625 & 0.375 \\ 0.375 & 0.625 \end{bmatrix}$$

$$P^4 = P^2 \cdot P^2 = \begin{bmatrix} 0.625 & 0.375 \\ 0.375 & 0.625 \end{bmatrix} \begin{bmatrix} 0.625 & 0.375 \\ 0.375 & 0.625 \end{bmatrix}$$

$$= \begin{bmatrix} 0.53 & 0.47 \\ 0.47 & 0.53 \end{bmatrix}$$

The required probability = 0.53
= 53%

Ans:-

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Sampling!

Chor-o

Sampling:

⇒ It's a technique for selecting representative sample from a population to study its characteristics.

Formulas:—

1. Estimate of sample means, $\bar{x} = \frac{1}{n} \sum x$.

2. Estimate of sample variance, $s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$

3. u of Variance of sample means, $V(\bar{x}) = \frac{N-n}{Nn} \times s^2$.

4. u Standard error of sample means, $s.e(\bar{x}) = \sqrt{V(\bar{x})}$.

5. u of total population, $\hat{X} = N\bar{x}$.

6. u of Variance of the estimate of total population.

$$V(\hat{X}) = N^2 V(\bar{x})$$

$$s.e(\hat{X}) = \sqrt{V(\hat{X})}$$

Where,

N = No. of population units.

n = No. of sample units.

7 Estimate of variance of proportion:-

$$V(P) = \frac{N-n}{N(n-1)} \cdot pqr \quad [q = 1-p]$$

8 Sample size: $n = \frac{Z^2 Pq}{d^2} \quad / \quad n = \frac{Z^2 V(\bar{x})}{d^2}$

9 Estimate the proportion of population:-

$$p = \frac{A}{N}$$

10 Estimate the proportion of sample:-

$$P = \frac{a}{n}$$

11 Estimate % Confidence limits for mean:-

$$\bar{M}_L = \bar{x} - t_{n-1} \cdot s.e(\bar{x})$$

$$\bar{M}_U = \bar{x} + t_{n-1} \cdot s.e(\bar{x})$$

where:-
N =

A = No. of units in population.

a = No. of units in sample.

List of formulas		
$v(\bar{x}) = \frac{N-n}{Nn} s^2$	$v(p) = \frac{N-n}{N(n-1)} pq$	$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$
$v(\hat{X}) = N^2 v(\bar{x})$	$s.e.(p) = \sqrt{v(p)}$	$\bar{x} \pm t_{n-1} * s.e.(\bar{x})$
$s.e.(\bar{x}) = \sqrt{v(\bar{x})}$	$\hat{X} = N\bar{x}$	$p = \frac{a}{n}$
$n = \frac{z^2 v(X)}{d^2}$	$n = \frac{z^2 pq}{d^2}$	$s.e.(\hat{X}) = \sqrt{v(\hat{X})}$
$P(T > t) = e^{-\lambda t}$	$P(T < t) = 1 - e^{-\lambda t}$	$P(t_1 < T < t_2) = e^{-\lambda t_1} - e^{-\lambda t_2}$
$z = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}}$	$t = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{n-1}$	$\chi^2 = \sum \sum \frac{O_{ij}^2}{E_{ij}} - n \sim \chi_{(r-1)(c-1)}^2$ $E_{ij} = \frac{R_i \times C_j}{n}$
$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$	$z = \frac{p - p_0}{\sqrt{\frac{p_0 Q_0}{n}}}$	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{(n_1-1)+(n_2-1)}$
$p = \frac{a_1 + a_2}{n_1 + n_2}$	$r = \frac{SP(xy)}{\sqrt{SS(x) SS(y)}}$	$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1-1) + (n_2-1)}$
$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$	$SS(y) = \sum y^2 - \frac{(\sum y)^2}{n}$	$\chi^2 = \sum \frac{O_i^2}{E_i} - n \sim \chi_{(k-1)}^2, E_i = \frac{n}{k}$
$SS(x) = \sum x^2 - \frac{(\sum x)^2}{n}$	$SP(xy) = \sum xy - \frac{\sum x \sum y}{n}$	$s^2 = \frac{ss(y) - b sp(xy)}{n-2}$
$\hat{y} = a + bx$	$t = \frac{b}{\sqrt{\frac{s^2}{SS(x)}}}$	$b = \frac{SP(xy)}{SS(x)}$

Random number		t Table [$\alpha = 0.05$]				χ^2 Table [$\alpha = 0.05$]	
		df	value	df	value	df	value
12212	01885					1	3.841
12294	10179	1	12.706	11	2.201	2	5.991
11584	17594	2	4.303	12	2.179	3	7.815
16986	15206	3	3.182	13	2.160	4	9.488
16141	09232	4	2.776	14	2.145	5	11.071
09294	41214	5	2.571	15	2.131	$z_{0.05} = 1.96$	
12584	54259	6	2.447	16	2.120		
16986	20888	7	2.365	17	2.110		
19278	16278	8	2.306	18	2.101		
10553	01177	9	2.262	19	2.093		
09876	16141	10	2.228	20	2.086		

Table of Random Numbers

36185, 46132, 31841, 84180, 165005, 55432,
11032, 100051, 88857, 700534, 432157, 105307,
205555, 3078254, 0000000, 173527, 273356, 210010

Problem 9.1: Number of pharmacies in various regions of a city are, X : 12, 20, 5, 25, 10, 35, 8, 15, 20, 13, 20, 18, 8, 9, 24, 25, 15, 30, 18, 22, 25, 17, 30, 25, 18, 20, 22, 20. Select a random sample of 6 regions by (i) simple random sampling, (ii) systematic sampling, (iii) circular systematic sampling.

1. Estimate mean number of pharmacies per region.
2. Estimate total number of pharmacies in the city.
3. Estimate standard error of estimated mean number of pharmacies.
4. Estimate standard error of estimated total number of pharmacies.
5. Find 95% confidence interval for mean number of pharmacies.
6. Suggest a sample of size n to estimate a population mean with margin of error 0.5 at 95% level of confidence, where variance of the population observations is 6.35.
7. Estimate the proportion of regions in which there are less than 18 pharmacies.
8. Estimate the variance of the estimated proportion.
9. Find sample size n to estimate proportion 0.75 with margin of error 0.2 at 95% confidence.

Observations (x)	12	20	5	25	10	35	8	15	20	13	20	18	8	9
Serial Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Observations (x)	24	25	15	30	18	22	25	17	30	25	18	20	22	20
Serial Number	15	16	17	18	19	20	21	22	23	24	25	26	27	28

(i)

$\therefore N = 28$ // No. of total data
 $n = 6$ // No. of sample data.

Random No.	16	11	10	19	09	17
Pharmacies of selected R.	25	20	13	18	20	15

(ii)

Let, $N = nk$

$$\therefore k^{\text{th}} = \frac{N}{n} = \frac{28}{6} = 4.7 \sim 5^{\text{th}} \text{ obs}$$

R.N.	1	6	11	16	22	26
Pharmacies of selected R.	12	35	20	25	25	20

(iii)

Also,

$$N = nk$$

$$\therefore k = \frac{N}{n} = 28/6 = 4.7 \sim 5$$

R.N	10	15	20	25	2	7
P.O.S.R	13	24	22	18	20	8

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①

we know:-

$$\bar{x} = \frac{1}{n} \sum x$$

$$= \frac{1}{6} (25 + 20 + 13 + 18 + 15 + 20)$$

$$= \frac{1}{6} \times 111$$

$$= 18.5$$

here,
 $n=6$

②

we know:-

$$\hat{X} = N \bar{x}$$

$$= 28 \times 18.5$$

$$= 518$$

here,

$$N = 28$$

$$\bar{x} = 18.5$$

if $\hat{X} = 518.00$ also

Then $\hat{X} = 518$

③

we know:-

$$S.E.(\bar{x}) = \sqrt{V(\bar{x})} \quad \text{--- (I)}$$

$$V(\bar{x}) = \frac{N-n}{Nn} s^2 \quad \text{--- (II)}$$

$$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

$$\therefore \sum x^2 = (25)^2 + (20)^2 + (13)^2 + (18)^2 + (15)^2 + (20)^2$$
$$= 2143$$

$$\therefore s^2 = \frac{1}{6-1} \left[2143 - \frac{(111)^2}{6} \right]$$
$$= 17.9$$

Now, put the value in eq (10) $\Rightarrow$

$$V(\bar{x}) = \frac{N-n}{Nn} \times s^2$$

$$= \frac{28-6}{28 \times 6} \times (17.9)$$

$$= 2.344.$$

Again, put the values in eq (1) :-

$$\therefore \text{s.e.}(\bar{x}) = \sqrt{V(\bar{x})}$$

$$= \sqrt{2.344}$$

$$= 1.531$$

(4)we know:-

$$s.e(\hat{X}) = \sqrt{V(\hat{X})} \text{ --- (1)}$$

$$\begin{aligned} V(\hat{X}) &= N^2 V(\bar{x}) \\ &= 28^2 \times 0.344 \\ &= 1837.60 \end{aligned}$$

Now put the value in eq (1)

$$\begin{aligned} \therefore s.e(\hat{X}) &= \sqrt{1837.60} \\ &= 42.86 \end{aligned}$$

Ans:-(5)we know:-

$$\bar{x}_L = \bar{x} - t_{n-1} \cdot s.e.(\bar{x}) \text{ --- (1)}$$

$$\bar{x}_U = \bar{x} + t_{n-1} \cdot s.e.(\bar{x}) \text{ --- (2)}$$

 $\therefore$ Lower Limit:-

$$\begin{aligned} \bar{x}_L &= 18.5 \times 1 - t_{6-1} \times 1.531 \\ &= 18.5 - t_5 \times 1.531 \\ &= 18.5 - 2.571 \times 1.531 \\ &= 14.56 \end{aligned}$$

here,

$$\bar{x} = 18.5$$

$$n = 6$$

$$s.e(\bar{x}) = 1.531$$

$$t_5 = 2.571$$

Upper Limit

$$\begin{aligned}\bar{x}_u &= \bar{x} + t_{n-1} \cdot se(\bar{x}) \\ &= 18.5 + 1.96 \times 1.531 \\ &= 18.5 + 2.971 \\ &= 22.43\end{aligned}$$

$$\therefore [14.57, 22.43]$$

Ans

In given

$$V(x) = 6.35$$

$$d = 0.5$$

$$Z_{0.05} = 1.96$$

we know

$$n = \frac{Z^2 \times V(x)}{d^2}$$

$$= \frac{(1.96)^2 \times 6.35}{(0.5)^2}$$

$$= 97.57$$

$$\approx 98$$

(7)To given:—less than 18 pharmacies from the sample are7, 13, 15 $\therefore a = 2 = \text{No. of units in sample/region.}$ $\& n = 6$ $\therefore$ we know:—

$$p = \frac{a}{n} = \frac{2}{6} = 0.33$$

Ans:—(8)we know:—Variance of proportion:—

$$V(p) = \frac{N-n}{N(n-1)} \times pq \quad [q = 1-p]$$

$$= \frac{28-6}{28(6-1)} \times (0.33) \times (1-0.33)$$

$$= 0.034$$

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Q

Is given: _____

$$P = 0.75$$

$$\therefore q = 1 - p = 1 - 0.75 = 0.25$$

$$Z = 1.96$$

$$d = 2$$

we know: _____

$$n = \frac{Z^2 pq}{d^2}$$

$$= \frac{(1.96)^2 \times 0.75 \times 0.25}{0.2^2}$$

$$= 18.0075$$

$$\approx 18$$

Ans:-

Unbiasedness of Mean & Variance:—

Let

$x: 2, 4, 6$

Now

We can select a sample $n = 2$.

Case: 1:—
mon

Sample mean unbiased

if Sample mean ($E(\bar{x})$) = population mean ($\bar{x}$)

Case: 2:—
mon

Sample Variance Unbiased:—

if Sample Variance ($E(s^2)$) = Population Variance (s^2)

∴ Example:—
mon

∴ population mean

$$\begin{aligned}\bar{x} &= \frac{1}{N} \sum x \\ &= \frac{1}{2+4+6} (2+4+6) \\ &= 12/3 = 4\end{aligned}$$

here
 $N = 3$

∴ population Variance

$$s^2 = \frac{1}{N-1} \left(\sum x^2 - \frac{(\sum x)^2}{N} \right) \quad \text{--- (1)}$$

$$\therefore \sum x^2 = 2^2 + 4^2 + 6^2 = 56$$

$$\sum x = 2 + 4 + 6 = 12$$

put them into the eq (1) =

$$\therefore s^2 = \frac{1}{3-1} \left(56 - \frac{12^2}{3} \right) = 4$$

Note: Since we are select 2 sample. So the possible
No. of Sample Combinations ${}^3C_2 = 3$

$\therefore$ Combinations of sample are:-

Comb: 1:- 2, 4

$$\therefore \bar{x}_1 = \frac{1}{n} \sum x = \frac{1}{2} (2+4) = 3$$

$$\therefore s_1^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{2-1} \left\{ (2^2+4^2) - \frac{(2+4)^2}{2} \right\} = 2$$

Comb: 2:- 2, 6

$$\therefore \bar{x}_2 = \frac{1}{n} \sum x = \frac{1}{2} (2+6) = 4$$

$$\therefore s_2^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{2-1} \left\{ (2^2+6^2) - \frac{(2+6)^2}{2} \right\} = 8$$

Comb: 3:- 4, 6

$$\therefore \bar{x}_3 = \frac{1}{n} \sum x = \frac{1}{2} (4+6) = 5$$

$$\therefore s_3^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{2-1} \left\{ (4^2+6^2) - \frac{(4+6)^2}{2} \right\} = 2$$

$$\therefore E(\bar{x}) = \frac{1}{3} (3+4+5) = 4 = \bar{X}$$

$$\therefore E(s^2) = \frac{1}{3} (2+8+2) = 4 = s^2$$

$\therefore$ Sample mean & Sample variance
are Unbiased.

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9.4 The number of signals received in a server in different days are, X : 5, 8, 7, 10, 7, 6, 9, 11,

4, 2, 7, 7, 12, 9, 11, 3, 7, 8, 5, 6, 7, 6, 9, 11, 4. Select 4 days by **systematic** sampling.

a) Estimate **total number of signals received per day** along with its estimated standard error.

b) Estimate the **proportion** of days in which **less than 8** signals are received.

→ In given:

$$N=25, n=4$$

$$\therefore k = \frac{N}{n} = \frac{25}{4} = 6.25 \approx 6 \quad [\because \text{Systematic sampling}]$$

Now,

Selected Sample days	3	9	15	21
Corresponding values in	7	4	11	7

(a)

we know:

$$\bar{x} = \frac{1}{n} \sum x = \frac{1}{4} \times (7+4+11+7) = 7.25$$

$$\therefore \hat{X} = N \times \bar{x} = 25 \times 7.25 = 181.25$$

(b)

In given,

Less than 8 signals are: 7, 4, 7

$$\therefore a=3$$

$$n=4$$

$$\therefore p = \frac{a}{n} = \frac{3}{4} = 0.75$$

Ans:-

(a)we know—

$$s.e(\bar{x}) = \sqrt{V(\bar{x})} \quad \text{--- (i)}$$

$$V(\bar{x}) = \frac{N-n}{Nn} \times s^2 \quad \text{--- (ii)}$$

$$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

$$= \frac{1}{4-1} \left[\sum x^2 - \frac{(\sum x)^2}{4} \right] \quad \text{--- (iii)}$$

$$\therefore \sum x^2 = 7^2 + 4^2 + 11^2 + 7^2 = 235$$

$$\sum x = 7 + 4 + 11 + 7 = 29$$

Now, put them into the eq (iii):

$$\therefore s^2 = \frac{1}{3} \left[235 - \frac{29^2}{4} \right]$$

$$= 8.25$$

Now, put the value in eq (ii):

$$V(\bar{x}) = \frac{25-4}{25 \times 4} \times s^2$$

$$= \frac{21}{100} \times 8.25$$

$$= 1.7325$$

Now, from eq (i):

$$s.e(\bar{x}) = \sqrt{1.7325} = 1.316$$

Ans:

9.5 The following are the number of faded out signals sent from a station in different days:

X: 4, 3, 0, 2, 6, 7, 4, 3, 2, 0, 1, 0, 3, 0, 6, 8, 0, 1, 4, 3, 2, 6, 3, 7, 5, 8, 0, 2, 3, 5.

Select a random sample of 5 days by simple random sampling method and estimate total number of faded out signals along with its estimated standard error

⇒ It is given:-

$$N = 30$$

$$n = 5$$

Since, Random Sampling:-

Random No.	3	10	15	21	25
values(x):-	0	0	6	2	5

(i)

we know:-

$$\bar{x} = \frac{1}{n} \sum x = \frac{1}{5} (0 + 0 + 6 + 2 + 5) = 2.6$$

$$\therefore \hat{X} = N \cdot \bar{x} = 30 \times 2.6 = 78$$

(ii)

$$\therefore \sum x^2 = 0^2 + 0^2 + 6^2 + 2^2 + 5^2 = 65$$

$$\therefore \sum x = 0 + 0 + 6 + 2 + 5 = 13$$

$$\therefore s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \frac{1}{5-1} \left(65 - \frac{(13)^2}{5} \right) = 7.8$$

$$\therefore V(\bar{x}) = \frac{N-n}{Nn} \times s^2 = \frac{30-5}{30 \times 5} \times 7.8 = 1.3$$

$$\therefore S.E(\bar{x}) = \sqrt{V(\bar{x})} = \sqrt{1.3} = 1.14$$

$$\therefore Se(\hat{X}) = N \cdot Se(\bar{x}) = 30 \times 1.14 = 34.2$$

9.6 Suggest a sample size n to estimate a proportion 0.45 of the number of days in which more than 10 signals are faded, with margin of error 0.1 at 95% confidence.

⇒ Is given:-

$$p = 0.45$$

$$\therefore q = 1 - p = 1 - 0.45 = 0.55$$

$$z = 1.96$$

$$d = 0.1$$

we know:-

$$n = \frac{z^2 \times p \times q}{d^2}$$

$$= \frac{(1.96)^2 \times 0.45 \times 0.55}{(0.1)^2}$$

$$= 95.0796$$

$$\approx 96$$

Ans:-

9.7 The number of mails received in a server at Bashundhara residential area in different days are:

X: 10, 7, 6, 9, 11, 4, 2, 7, 7, 9, 11, 45, 8, 7, 10, 7, 6, 9, 11, 4, 2, 7, 7.

Select 4 days by simple random sampling. Estimate mean number of emails received per day along with its estimated standard error.

⇒ To given: _____

$$N = 23$$

$$n = 4$$

Since Random Sampling.

Random No.	2	5	12	19
corr. values(x)	7	11	45	11

i

we know,

$$\bar{x} = \frac{1}{n} \sum x = \frac{1}{4} (7 + 11 + 45 + 11) = 18.5$$

ii

we know,

$$\sum x^2 = 7^2 + 11^2 + 45^2 + 11^2 = 2316$$

$$\sum x = 7 + 11 + 45 + 11 = 74$$

$$\therefore s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \frac{1}{4-1} \left(2316 - \frac{74^2}{4} \right)$$

$$= 315.667$$

$$\therefore V(\bar{x}) = \frac{N-n}{Nn} \times s^2 = \frac{23-4}{23 \times 4} \times 315.67$$

$$= 65.102$$

$$\therefore s.e(\bar{x}) = \sqrt{V(\bar{x})} = \sqrt{65.102} = 8.074$$

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9.8 Suggest a sample size n to estimate a proportion 0.3 of the number of days in which more than 10 signals are faded, with margin of error 0.05 at 95% confidence.

⇒ Is given:

$$p = 0.3$$

$$\therefore q = 1 - p = 1 - 0.3 = 0.7$$

$$Z = 1.96$$

$$d = 0.05$$

we know:

$$n = \frac{Z^2 \cdot p \cdot q}{d^2} = \frac{(1.96)^2 \times 0.7 \times 0.3}{(0.05)^2}$$

$$= 322.6044$$

$$\approx 323$$

Ans:

9.9 The number of noisy bits produced by an electronic device in different attempts is given as,

X: 5, 8, 7, 10, 7, 6, 9, 11, 4, 2, 7, 7, 12, 9, 11, 3, 7, 8, 5, 6.

Select a random sample of 5 attempts by circular systematic sampling and find 95% confidence interval for the average of noisy bits.

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⇒ To given:

$$N = 20$$

$$n = 5$$

$$\therefore k = \frac{N}{n} = \frac{20}{5} = 4 \quad [\because \text{Systematic Sampling}]$$

Now:

Selected Sample device	3	7	11	15	19
Corresp. Values	7	9	7	11	5

∴ Lower Limit:

$$\bar{x}_L = \bar{x} - \sqrt{n-1} \cdot se(\bar{x}) \quad \text{--- (i)}$$

& Upper Limit:

$$\bar{x}_U = \bar{x} + \sqrt{n-1} \cdot se(\bar{x}) \quad \text{--- (ii)}$$

Now:

we know:

$$\bar{x} = \frac{1}{n} \sum x = \frac{1}{5} (7+9+7+11+5) = 7.8$$

$$\sum x^2 = 7^2 + 9^2 + 7^2 + 11^2 + 5^2 = 325$$

$$\sum x = 7+9+7+11+5 = 39$$

$$\therefore s^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{5-1} \left(325 - \frac{(39)^2}{5} \right) = 5.2$$

$$\therefore se(\bar{x}) = \sqrt{\frac{N-n}{N \cdot n} s^2}$$

$$= \sqrt{\frac{20-5}{20 \times 5} \times 5.2} = 0.882$$

Now, put the values in eq (i) $\Rightarrow$

$$\begin{aligned}\therefore \bar{x}_L &= 7.8 - t_{5-1} \cdot (0.883) \quad [\because t_4 = 2.776] \\ &= 7.8 - t_4 \cdot (0.883) \\ &= 7.8 - 2.776 \cdot (0.883) \\ &= 5.34\end{aligned}$$

Now, put those values in eq (ii) $\Rightarrow$

$$\begin{aligned}\bar{x}_U &= 7.8 + t_{5-1} \cdot (0.883) \\ &= 7.8 + t_4 \cdot (0.883) \\ &= 7.8 + 2.776 \cdot (0.883) \\ &= 10.25\end{aligned}$$

$\therefore$ The interval $= [5.34, 10.25]$

Ans:-

Q:- The no. of noising bit produced by an electric device in different attempts is given as

20: 5, 8, 7, 10, 7, 6, 9, 11, 4, 2, 7, 7, 12, 9, 11,
3, 7, 8, 5, 6.

Select a random sample of 5 attempts using the last digit of given number table and find the 95% confidence interval for average noising bits

t	0.05
4	0.182
5	0.2776
6	0.571

No table	
6366	1088
0025	1014
5818	0804
4609	0622
2821	0311

⇒ In given:

$$N = 20$$

$$n = 5$$

∴ Random Sampling:

Random No.	18	00	14	04	11
Corresponding Value	8	4	9	10	7

⇒ We know: —

$$\bar{x}_L = \bar{x} - t_{n-1} \cdot se(\bar{x}) \quad \text{--- (i)}$$

$$\bar{x}_U = \bar{x} + t_{n-1} \cdot se(\bar{x}) \quad \text{--- (ii)}$$

Now,

$$\bar{x} = \frac{1}{n} \sum x = \frac{1}{5} (8+4+0+10+7) = 7.6$$

$$\therefore \sum x^2 = 8^2 + 4^2 + 0^2 + 10^2 + 7^2 = 310$$

$$\sum x = 8 + 4 + 0 + 10 + 7 = 38$$

$$\therefore s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

$$= \frac{1}{5-1} \left[310 - \frac{38^2}{5} \right]$$

$$= 5.3$$

$$\therefore se(\bar{x}) = \sqrt{\frac{N-n}{Nn} s^2}$$

$$= \sqrt{\frac{20-5}{20 \times 5} \times (5.3)}$$

$$= 0.89$$

Now, put the values in eq. (i), (ii):—

$$\therefore \bar{x}_L = 7.6 - t_{4} \times (0.89)$$

$$= 7.6 - 3.182 \times 0.89$$

$$= 4.768$$

$$\therefore \bar{x}_U = 7.6 + 3.182 \times 0.89$$

$$= 10.49$$

∴ The Interval is —

$$[4.768, 10.49]$$

Ans.

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#HYPOTHESIS

Ch^o 10!

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Hypothesis test:-

⇒ It's a formal way to make a decision based on Statistical analysis.

There are two types:-

1. Null hypothesis.

2. Alternative hypothesis.

Null Hypothesis: (H_0)

⇒ No Difference exists between parameters.

Denote by " H_0 "

Example:-

$$H_0 : \mu_1 = \mu_2$$

Alternative Hypothesis: (H_1)

⇒ Opposite of the null hypothesis. Denote by " H_1 "

Example:-

$$H_1 : \mu_1 \neq \mu_2$$

OR $H_1 : \mu_1 > \mu_2$

OR $H_1 : \mu_2 < \mu_1$

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Formulas:-

✶ Test for One Variable

⇒ T-test ($n < 30, \sigma^2 = \text{Unknown}$)

$$\Rightarrow t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

$$\Rightarrow \bar{x} = \frac{\sum x_i}{n}$$

$$\Rightarrow s^2 = \frac{1}{n-1} \left[\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right]$$

⇒ Z-test ($n \geq 30, \sigma^2 = \text{known}$)

$$\Rightarrow z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim N(0,1)$$

⇒ Z-test (proportion)

$$\Rightarrow z = \frac{p - p_0}{\sqrt{\frac{p_0 q_0}{n}}} \sim N(0,1)$$

$$\Rightarrow p = \frac{a}{n}$$

$$\Rightarrow p_0 = \frac{A}{N}$$

$$\Rightarrow q_0 = 1 - p_0$$

✶ Test for two Variable:-

$$\Rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{(n_1-1) + (n_2-1)}$$

$$\Rightarrow s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1-1) + (n_2-1)}$$

$$\Rightarrow \bar{x}_1 = \frac{1}{n_1} \sum x_{1i} ; \Rightarrow \bar{x}_2 = \frac{1}{n_2} \sum x_{2i}$$

$$\Rightarrow z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)}} \sim N(0,1)$$

$$\Rightarrow z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0,1)$$

$$\Rightarrow p = \frac{a_1 + a_2}{n_1 + n_2}$$

$$\Rightarrow q = 1 - p$$

Test of independence

$$\chi^2 = \sum \sum \frac{O_{ij}^2}{E_{ij}} - n \sim \chi^2_{(r-1)(c-1)}$$

row column.

here, O_{ij} = Observation of "ith" row & "jth" column. E_{ij} = Expected frequency corresponding to "ith" row & "jth" column.# problems 10.5:-

⇒ For a Sample size 36, $\sum x = 761.6$, $\sum x^2 = 16125.5$
In the population mean 21.

⇒ We need to test:-

$$H_0: \mu = \mu_0 = 21 \quad \text{Vs} \quad H_1: \mu \neq \mu_0$$

We know:-

$$\begin{aligned} Z &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ &= \frac{21.156 - 21}{\sqrt{0.38}/\sqrt{36}} \\ &= 1.518 \end{aligned}$$

We know:-

$$\begin{aligned} \bar{x} &= \frac{\sum x}{n} = \frac{761.6}{36} = 21.156 \\ \therefore s^2 &= \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] \\ &= \frac{1}{35} \left[16125.5 - \frac{(761.6)^2}{36} \right] \\ &= 0.38 \end{aligned}$$

In given:-

$$\begin{aligned} n &= 36 > 30 \rightarrow \text{7 tent} \\ \sum x^2 &= 16125.5 \\ \sum x &= 761.6 \\ \therefore \mu_0 &= 21 \end{aligned}$$

$$\therefore |Z| = 1.06$$

$$\Rightarrow |1.518| \neq 1.06$$

$$\Rightarrow |1.518| < 1.06$$

 $\therefore H_0 = \text{Accepted}$

So, Yes population mean is 21

problem: 10.6:-

⇒ A company claims that its batteries have a mean life of 100 hr. You try to verify this for sample of size 21 with mean 97 hours, and variance 9 hr.

⇒ We need to test:-

$$H_0: \mu = \mu_0 = 100 \text{ Vs } H_1: \mu \neq \mu_0$$

We know:-

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

$$= \frac{97 - 100}{\sqrt{9}/\sqrt{21}}$$

$$= -4.58$$

Is given:-

$$n = 21 \text{ (30) } t_{20}$$

$$s^2 = 9$$

$$\bar{x} = 97$$

$$\mu_0 = 100$$

$$t_{20} = 2.086$$

$$\therefore |t| < t_{n-1}$$

$$\Rightarrow |-4.58| < t_{21-1}$$

$$\Rightarrow |-4.58| < t_{20}$$

$$\Rightarrow |-4.58| < 2.086$$

∴ Rejected

∴ The Batteries doesn't have the mean life of 100 hr.

problem: 10.8:-

⇒ A researcher claims that 10-year old children watch 6.6 hours of TV daily on average. You try to verify this for a sample of size 100 with mean 6.1 hours and standard deviation 2.5 hr.

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⇒ We need to test:-

$$H_0: \mu = \mu_0 = 6.6 \quad \text{Vs} \quad H_1: \mu \neq \mu_0$$

$$\therefore |z| < 1.96$$

$$\Rightarrow |z| < 1.96$$

⇒ (Rejected)

$$\begin{aligned} \text{We know:-} \\ z &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ &= \frac{6.1 - 6.6}{2.5/\sqrt{100}} \\ &= -2 \end{aligned}$$

Is given:-

$$n = 100 > 30 \rightarrow z\text{-test}$$

$$\bar{x} = 6.1$$

$$s = 2.5 \text{ hr}$$

$$\mu_0 = 6.6 \text{ hr}$$

∴ The claim is not True

Ans

problem: 10.7:-

⇒ Out of 25 students 8 are female. In the overall proportion of female students 0.40 in A.I.U.B.

⇒ We need to test:-

$$H_0: p = p_0 = 0.40 \quad \text{Vs} \quad H_1: p \neq p_0$$

$$\therefore |z| < 1.96$$

$$\Rightarrow |z| < 1.96$$

∴ Accepted

So the claim is True.

$$\begin{aligned} \text{We know:-} \\ p &= \frac{a}{n} = \frac{8}{25} = 0.32 \\ q_0 &= 1 - p_0 = 1 - 0.4 = 0.6 \\ \therefore z &= \frac{p - p_0}{\sqrt{p_0 q_0 / n}} = \frac{0.32 - 0.4}{\sqrt{\frac{0.6 \times 0.4}{25}}} \\ &= -0.82 \end{aligned}$$

Is given:-

$$a = 8$$

$$n = 25$$

$$p_0 = 0.40$$

Example 10.4: A sample of 15 students is selected from a group of 100 students and their grade in SSC examination is recorded as follows:

Students	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Grade	B	C	A	D	B	C	D	A	B	C	D	B	C	C	D

Do you think that 10% students get grade A?

⇒ We need to test:-

$$H_0: P = P_0 = 10\% = 0.1 \quad \text{Vs} \quad H_1: P \neq P_0$$

we know:-

$$P = \frac{a}{n} = \frac{2}{15} = 0.133$$

$$Q_0 = 1 - P_0 = 1 - 0.1 = 0.9$$

$$\therefore Z = \frac{P - P_0}{\sqrt{\frac{P_0 Q_0}{n}}}$$

$$= \sqrt{\frac{0.133 - 0.1}{\frac{0.1 \times 0.9}{15}}}$$

$$= 0.382$$

Ans:-

It is given:-

$$a = 2$$

$$n = 15$$

$$P_0 = 0.1$$

$$\therefore |Z| < 1.96$$

$$\Rightarrow |0.382| < 1.96$$

$\therefore H_0 = \text{Accepted}$

So Yes, 10%.

Students get A grade.

Example 10.5: Is the severity of the drug problem in high school the same for boys and girls? 85 boys and 70 girls were questioned and 34 of the boys and 14 of the girls admitted to having tried some sort of drug. What can be concluded at the 0.05 level?

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⇒ we need to test!

$$H_0 : P_B = P_G \quad \text{Vs} \quad P_B \neq P_G$$

⇒ we know!

$$\begin{aligned} Z &= \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \\ &= \frac{0.4 - 0.2}{\sqrt{0.31 \times 0.69 \left(\frac{1}{85} + \frac{1}{70} \right)}} \\ &= 2.68 \end{aligned}$$

we know!

$$\begin{aligned} P_1 = P_B &= \frac{a_1}{n_1} = \frac{34}{85} = 0.4 \\ P_2 = P_G &= \frac{a_2}{n_2} = \frac{14}{70} = 0.2 \\ \therefore P &= \frac{a_1 + a_2}{n_1 + n_2} = \frac{34 + 14}{85 + 70} \\ Q &= 1 - P = 1 - 0.31 \\ &= 0.69 \end{aligned}$$

Is given!

$$\begin{aligned} a_1 = a_B &= 34 \\ a_2 = a_G &= 14 \\ n_1 = n_B &= 85 \\ n_2 = n_G &= 70 \end{aligned}$$

$$\therefore |Z| < 1.96$$

$$\Rightarrow 2.68 < 1.96$$

∴ Rejected

so Severity of the drug proportion are not same!

Ans!

Example 10.2: The number of computer graduates coming out from two different universities A and B are employed in different organizations to do job related to computer.

University	Number of graduates employed in computer related job
A	$x_1 : 18, 16, 15, 20, 18, 15, 12; n_1 = 7$
B	$x_2 : 20, 14, 12, 22, 16, 14, 15, 10, 12, 18, 10; n_2 = 11$

Do you think that the employment facility for both the universities is similar?

⇒ We need to test:—

$H_0: \mu_1 = \mu_2$ Vs $\mu_1 \neq \mu_2$

We know:—

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1-1) + (n_2-1)}$$

$$= \frac{6 \cdot (6.205) + 10 \cdot (15.364)}{6 \times 1 + 10 \times 1}$$

$$= 12.101$$

$$\therefore t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{16.20 - 14.92}{\sqrt{12.101 \left(\frac{1}{7} + \frac{1}{11} \right)}}$$

$$= 0.87$$

We know:—

$$\bar{x}_1 = \frac{1}{n_1} \sum x_1 = \frac{1}{7} (18 + 16 + 15 + 20 + 18 + 15 + 12)$$

$$= 16.20$$

$$\bar{x}_2 = \frac{1}{n_2} \sum x_2 = \left(\frac{1}{11} \right) (20 + 14 + \dots + 10)$$

$$= 14.92$$

$$s_1^2 = \frac{1}{n_1-1} \left(\sum x_1^2 - \frac{(\sum x_1)^2}{n_1} \right)$$

$$= \frac{1}{6} \left(1808 - \frac{(144)^2}{7} \right)$$

$$= 6.205$$

$$s_2^2 = \frac{1}{n_2-1} \left(\sum x_2^2 - \frac{(\sum x_2)^2}{n_2} \right)$$

$$= \frac{1}{10} \left(2560 - \frac{(163)^2}{11} \right)$$

$$= 15.364$$

It is given:—

$$n_1 = 7 < 30$$

$$n_2 = 11 \rightarrow \text{t-test}$$

$$\sum x_1 = 144$$

$$\sum x_1^2 = 1808$$

$$\sum x_2 = 163$$

$$\sum x_2^2 = 2560$$

$$\therefore |t| < t_{(n_1-1)(n_2-1)}$$

$$\Rightarrow |0.87| < t_{(7-1)(11-1)}$$

$$\Rightarrow |0.87| < t_{16}$$

$$\Rightarrow |0.87| < 2.12$$

∴ (Accepted)

So, Yes, The employment facility for both of them is similar.

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10.10 The information about daily temperature (in $^{\circ}$ Celsius) of two months are as:

Month 1	$n_1 = 31$	$\sum x_1 = 1032$	$s_1^2 = 1.41$
Month 2	$n_2 = 30$	$\sum x_2 = 1035$	$s_2^2 = 1.09$

Do you think that the temperature of both the months are similar?

$\Rightarrow$ We need to test:

$$H_0: \mu_1 = \mu_2 \quad \text{Vs} \quad H_1: \mu_1 \neq \mu_2$$

$\therefore$ Now,

$$\begin{aligned} \bar{z} &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \\ &= \frac{33.29 - 34.5}{\sqrt{\left(\frac{1.41}{31} + \frac{1.09}{30}\right)}} \\ &= -4.23 \end{aligned}$$

we know:

$$\begin{aligned} \bar{x}_1 &= \frac{1}{n_1} (\sum x_1) = \frac{1}{31} (1032) = 33.29 \\ \bar{x}_2 &= \frac{1}{n_2} (\sum x_2) = \frac{1}{30} (1035) = 34.5 \end{aligned}$$

Is given:

$$\begin{aligned} n_1 &= 30 + 1 = 31 \\ n_2 &= 30 \\ \sum x_1 &= 1032 \\ \sum x_2 &= 1035 \\ s_1^2 &= 1.41 \\ s_2^2 &= 1.09 \end{aligned}$$

$$\therefore |z| < 1.96$$

$$\Rightarrow |-4.23| < 1.96$$

(Rejected)

So, Temperature of both months are not similar.

Ans:-

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Example 10.7: 150 computer graduates are interviewed and are classified according to their result and job satisfaction. Do you think that the graduates with good result are satisfied with their job?

Result	Job satisfaction		Total
	Yes	No	
Good	22	58	80
Not good	20	50	70
Total	42	108	150

⇒ we need to test:-

H_0 : Job satisfaction does not depend on good result

H_1 : The good result and job satisfaction are associated.

We know:-

$$\chi^2 = \sum \sum \frac{O_{ij}^2}{E_{ij}} - n ; \sim \chi_{(r-1)(c-1)}^2$$

$$E_{ij} = \frac{R_i C_j}{n}$$

Result	Satisfied	Not Satisfied	Total
Good	22 $\frac{80 \times 42}{150} = 22.4$	58 $\frac{108 \times 80}{150} = 57.6$	80
Not Good	20 $\frac{42 \times 70}{150} = 19.6$	50 $\frac{108 \times 70}{150} = 50.4$	70
Total	42	108	150

here, $n = 150$

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$$\begin{aligned}\therefore \chi^2 &= \left(\frac{22^2}{22.4} + \frac{58^2}{57.6} + \frac{20^2}{10.6} + \frac{50^2}{50.4} \right) - n \\ &= 150.021 - 150 \\ &= 0.021\end{aligned}$$

$$\therefore |\chi^2| < \chi^2_{(n-1)(c-1)}$$

$$\Rightarrow |0.021| < \chi^2_{(2-1)(2-1)}$$

$$\Rightarrow |0.021| < \chi^2_1$$

$$\Rightarrow |0.021| < 3.84$$

$\therefore$ (Accepted)

So Yes, job satisfaction does not depend on good result.

Ans:-

Example 10.8: The following are number of emails of different organizations, where emails are classified according to local and foreign emails. The classified results are shown below:

Origin of emails	Emails of organizations (O_{ij})				
	O_1	O_2	O_3	O_4	Total
Local	8	22	25	20	75
Foreign	17	73	25	10	125
Total	25	95	50	30	200

Is there any association between origin of emails and organizations?

⇒ We need to test:—

H_0 : Origin of Emails doesnot depend on organization

H_1 : Origin of Emails depend on organization.

We know:—

$$\chi^2 = \sum \sum \frac{O_{ij}^2}{E_{ij}} - n \sim \chi^2_{(r-1)(c-1)}$$

$$E_{ij} = \frac{R_i C_j}{n}$$

origin of Emails	O_1	O_2	O_3	O_4	Total
Local	8 $\frac{25 \times 75}{200} = 9.375$	22 $\frac{25 \times 75}{200} = 9.375$	25 $\frac{50 \times 75}{200} = 18.75$	20 $\frac{30 \times 75}{200} = 11.25$	75
Foreign	17 $\frac{25 \times 125}{200} = 15.625$	73 $\frac{25 \times 125}{200} = 15.625$	25 $\frac{50 \times 125}{200} = 31.25$	10 $\frac{30 \times 125}{200} = 18.75$	125
Total	25	95	50	30	$n=200$

$$\therefore \chi^2 = \left(\frac{8^2}{9.375} + \frac{22^2}{9.375} + \frac{25^2}{18.75} + \frac{20^2}{11.25} + \frac{17^2}{15.625} + \frac{73^2}{15.625} + \frac{25^2}{31.25} + \frac{10^2}{18.75} \right) - 200$$

$$= 222.88 - 200 = 22.88$$

$$\therefore |\chi^2| < \chi^2_{(r-1)(c-1)}$$

$$\Rightarrow |22.88| < \chi^2_{(2-1)(4-1)}$$

$$\Rightarrow |22.88| < \chi^2_3$$

$$\Rightarrow |22.88| < 7.815$$

∴ Rejected

∴ Origin of Emails depends on Organization.

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10.1 Among 157 African-American men, the mean systolic blood pressure was 146 mm Hg with a standard deviation of 27. We wish to know if on the basis of these data, we may conclude that the mean systolic blood pressure for a population of African-American is 140.

⇒ we need to test that:—

$$H_0: \mu_0 = \mu = 140 \quad \text{Vs} \quad H_1: \mu_0 \neq \mu$$

∴

$$|z| < 1.96$$

$$|2.78| < 1.96$$

∴ Rejected

we know:—

$$\begin{aligned} z &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ &= \frac{146 - 140}{27/\sqrt{157}} \\ &= 2.78 \end{aligned}$$

is given:—

$$n = 157 > 30 \rightarrow z\text{-test}$$

$$\bar{x} = 146$$

$$\mu_0 = 140$$

$$s = 27$$

So, The mean Systolic blood pressure for a population of African-American is not 140.

10.2 Are the proportions of road accidents similar in various highways of Bangladesh?

Highways	1	2	3	4	Total
No. of road accidents (O_i)	50	42	32	82	206

⇒ we need to test —

H_0 : The proportions of road accidents similar across the all highways.

H_1 : The proportion of road accidents are not similar across the all highways.

⇒ we know —

$$E_i = \frac{n}{k} = \frac{206}{4} = 51.5$$

$$\therefore \chi^2 = \sum \frac{O_i^2}{E_i} - n$$

$$= \left[\frac{1}{51.5} (50^2 + 42^2 + 32^2 + 82^2) \right] - 206$$

$$= 233.24 - 206$$

$$= 27.24$$

Is given —

$$n = 206$$

$$k = 4$$

$$\therefore |\chi^2| < \chi^2_{(k-1)}$$

$$\Rightarrow |27.24| < \chi^2_3$$

$$\Rightarrow |27.24| < 7.81$$

(Rejected)

∴ The proportion of road accidents are not similar across the all highways.

Ans —

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#Correlation

Ch 17

Correlation

⇒ A measure of linear Relationship between two quantitative variables. Such as age & weight.

Types of correlation:

1. Positive Correlation.
2. Negative Correlation.
3. No (Zero) Correlation.

Correlation formulas: (r)

$$\Rightarrow r = \frac{\text{Cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}} = \frac{SP(X, Y)}{\sqrt{SS(X) \cdot SS(Y)}} = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sqrt{\left[\sum X^2 - \frac{(\sum X)^2}{n}\right] \left[\sum Y^2 - \frac{(\sum Y)^2}{n}\right]}}$$

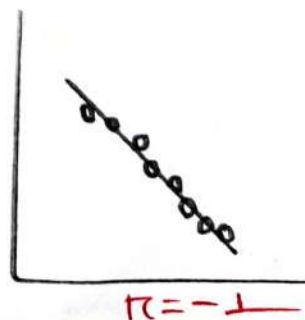
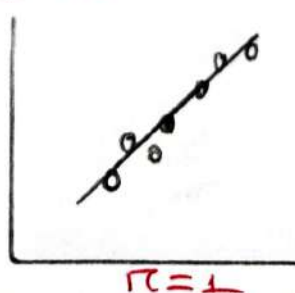
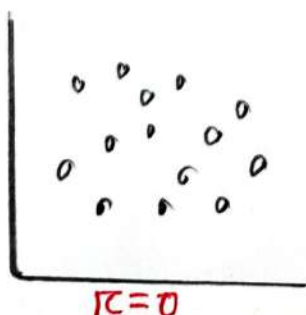
⇒ 1.0:- No/Zero Correlation $r = 0$

⇒ 2.0:- positive Correlation $r = 0 < r \leq 1$

⇒ 3.0:- Negative Correlation $r = -1 \leq r < 0$

⇒ Significance test:-

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$



Regression :-

$\Rightarrow Y = a + bx + c$

1. $\hat{Y} = a + bx$

2. $a = \bar{Y} - b\bar{x}$

3. $b = \frac{SP(XY)}{SS(X)}$

4. $\bar{Y} = \frac{\sum Y}{n}$

5. $\bar{x} = \frac{\sum x}{n}$

Significance test :-

$\Rightarrow t = \frac{b}{\sqrt{\frac{s^2}{SS(X)}}}$

$\Rightarrow s^2 = \frac{SS(Y) - bSP(XY)}{n-2}$

Examples :-

Age	25	30	35	30	32	40	45	40	36	35
BP	75	80	85	90	95	85	100	90	85	80

① Compute Correlation Coefficient

② Do you think BP increases significantly with the increase in age?

⇒

x	y	xy	x^2	y^2
25	75	1875	625	5625
30	80	2400	900	6400
35	85	2975	1225	7225
30	90	2700	900	8100
32	95	3040	1024	9025
40	85	3400	1600	7225
45	100	4500	2025	10000
40	90	3600	1600	8100
36	85	3060	1296	7225
35	80	2800	1225	6400
$\Sigma x = 348$	$\Sigma y = 865$	$\Sigma xy = 30350$	$\Sigma x^2 = 12420$	$\Sigma y^2 = 75325$

$n=10$

①

⇒ We know:-

$$r = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right] \left[\Sigma y^2 - \frac{(\Sigma y)^2}{n}\right]}}$$

$$= \frac{30350 - \frac{348 \times 865}{10}}{\sqrt{\left[12420 - \frac{348^2}{10}\right] \left[75325 - \frac{865^2}{10}\right]}}$$

$$= 0.62875 \dots$$

$$\approx 0.63$$

(6)

we need to test $H_0: \rho = 0 \rightarrow$ Not significant. $H_1: \rho = 1 \rightarrow$ significant.

$$\therefore |t| < t_{n-1}$$

$$\Rightarrow |2.20| < 1.0$$

$$\Rightarrow |2.20| < 2.306$$

 $\therefore H_0$ is accepted $\therefore$ we know:-

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$= \frac{0.63\sqrt{10-2}}{\sqrt{1-(0.63)^2}}$$

$$= 2.20$$

In given:-

$$n = 10$$

$$r = 0.63$$

so Age and BP

are not significantly increases.

Ans:-# problem 1:- $\Rightarrow$ show by example that $r=1$ $\Rightarrow$ let, $x = 1, 2, 3$; $y = 1, 2, 3$

$$\therefore \sum x = 6; \quad \sum y = 6$$

$$\sum xy = 14; \quad \sum x^2 = 14; \quad \sum y^2 = 14$$

$$\therefore r =$$

$$r = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}} = 1$$

$$\begin{aligned}
 \Rightarrow r &= \frac{14 - \frac{6 \times 6}{10}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}} \\
 &= \frac{14 - \frac{36}{10}}{\sqrt{\left(14 - \frac{6^2}{10}\right) \left(14 - \frac{6^2}{10}\right)}} \\
 &= 1
 \end{aligned}$$

Ans:-

problem 2:-

$\Rightarrow$ Show by example that $r = 1$

$\Rightarrow$ Let $x = 1, 2, 3$; $y = 2, 1, 1$

$$\therefore \sum x = 6 ; \sum y = 6$$

$$\sum xy = 10 ; \sum x^2 = 14 ; \sum y^2 = 14$$

$$\begin{aligned}
 \therefore r &= \frac{10 - \frac{6 \times 6}{3}}{\sqrt{\left(14 - \frac{6^2}{3}\right) \left(14 - \frac{6^2}{3}\right)}} \\
 &= 1
 \end{aligned}$$

$\therefore$ (Proved!)

Ans:-

Problem 11.2: The following are the data representing the number of ever born children (y) to different mothers having different levels of education (x in completed years of schooling):

x : 8, 4, 5, 10, 12, 8, 5, 10, 0, 6, 8, 5, 0

y : 2, 6, 5, 3, 1, 2, 5, 2, 7, 3, 4, 2, 5

- Fit a regression line of y on x .
- Estimate the number of children of a mother who complete 14 years of schooling.
- Test the significance of regression.

(a)

$\Rightarrow$ we know:-

Fitted line $\hat{y} = a + bx$ ——— (i)

$$a = \bar{y} - b\bar{x} \text{ ——— (ii)}$$

$$b = \frac{SP(xy)}{SS(x)} \text{ ——— (iii)}$$

$$\bar{y} = \frac{\sum y}{n} = \frac{2+6+5+3+1+\dots+2+5}{13} = \frac{47}{13}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{8+4+5+10+\dots+5+0}{13} = \frac{81}{13}$$

$$\begin{aligned} SS(x) &= \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \left[(8^2 + 4^2 + 5^2 + \dots + 5^2 + 0^2) - \frac{(81)^2}{13} \right] \\ &= \left[669 - \frac{81^2}{13} \right] \\ &= 158.31 \end{aligned}$$

$$\begin{aligned} SP(xy) &= \sum xy - \frac{\sum x \sum y}{n} = \left[(8 \times 2 + 4 \times 6 + \dots + 5 \times 0) - \frac{81 \times 47}{13} \right] \\ &= \left[228 - \frac{3807}{13} \right] = -64.85 \end{aligned}$$

Now, put the values in eq (ii):

$$b = \frac{SP(XY)}{SS(X)} = \frac{-64.85}{158.31} = -0.41$$

& Now, in eq (iii):

$$a = \bar{Y} - b\bar{X} = \frac{47}{13} - (-0.41) \frac{81}{13}$$

$$= 6.17$$

& Now, put the values in eq (i):

$$\hat{Y} = a + bX$$

$$= 6.17 - 0.41X$$

Ans:

(b)

⇒ It is given:

$$X = 14$$

$$\hat{Y} = 6.17 - 0.41X \longrightarrow [\text{From "a"}]$$

$$= 6.17 - 0.41 \times 14$$

$$= 0.42$$

⇒ We need to test: (c)

$$H_0: \beta = 0 \text{ Vs } H_1: \beta \neq 0$$

we know —

$$t = \frac{b}{\sqrt{\frac{s^2}{SS(x)}}} \quad \text{--- (1)}$$

$$\begin{aligned} s^2 &= \frac{SS(y) - b \cdot SP(xy)}{n-2} \\ &= \frac{41.08 - (-0.41)(-6485)}{13-2} \\ &= \frac{41.08 - 26.5885}{11} \\ &= 1.317 \end{aligned}$$

Now,

$$|t| < t_{n-2}$$

$$\Rightarrow |4.495| < t_{13-2}$$

$$\Rightarrow |4.495| < t_{11}$$

$$\Rightarrow |4.495| \nless 2.201$$

$H_0 = \text{Rejected}$

So, the regression is significant

Ans. —

put the values in (1) $\Rightarrow$

$$\begin{aligned} t &= \frac{-0.41}{\sqrt{\frac{1.317}{158.31}}} \\ &= 4.495 \end{aligned}$$

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problem 11.1

11.1 The following data are given for the inflation rate(x) and the corresponding lending rate(y)

x	y			
11.8	10.4			
12.5	16.5			
15.7	22.9			
19.2	26.6			
21.9	33.8			
23.3	42.8			

- Compute correlation coefficient.
- Do you think that lending rate increases significantly with the increase of inflation rate?
- Fit a regression line of y on x.
- Estimate the lending rate when the inflation rate will be 25.5.
- Test the significance of regression.

⇒ we know

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{(\sum x^2 - \frac{(\sum x)^2}{n})(\sum y^2 - \frac{(\sum y)^2}{n})}} \quad \text{--- (I)}$$

$$\sum x^2 = \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] \quad \text{--- (II)}$$

$$\sum y^2 = \left[\sum y^2 - \frac{(\sum y)^2}{n} \right] \quad \text{--- (III)}$$

Now

$$\sum x^2 = (11.8)^2 + (12.5)^2 + (15.7)^2 + \dots + (23.3)^2 = 1033.12$$

$$\sum x = 11.8 + 12.5 + 15.7 + \dots + (23.3) = 104.4$$

$$\sum y^2 = (10.4)^2 + (16.5)^2 + (22.9)^2 + \dots + (42.8)^2 = 4586.66$$

$$\sum y = 10.4 + 16.5 + 22.9 + \dots + 42.8 = 158$$

$$\sum xy = (11.8 \times 10.4) + [(12.5 \times 16.5)] + \dots + (23.3 \times 42.8) = 2036.68$$

Now put the values in eq (10), (11) & then (1)

$$\therefore SS(Y) = \left(\sum Y^2 - \frac{(\sum Y)^2}{n} \right)$$

$$= \left(4586.66 - \frac{(158)^2}{6} \right)$$

$$= 425.00$$

is given:-
 $n=6$

$$\therefore SS(X) = \left(\sum X^2 - \frac{(\sum X)^2}{n} \right)$$

$$= \left(1093.12 - \frac{(104.4)^2}{6} \right)$$

$$= 116.56$$

$$\therefore r = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sqrt{SS(X) \cdot SS(Y)}}$$

$$= \frac{2936.68 - \frac{104.4 \times 158}{6}}{\sqrt{(116.56) \times (425.00)}}$$

$$= 0.841$$

(6)

$\Rightarrow$ we need to test:-

$H_0: \rho = 0 \rightarrow$ Not increases significantly

$H_1: \rho = 1 \rightarrow$ Increases significantly

we know:-

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

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$$t = \frac{0.841 \times \sqrt{6-2}}{\sqrt{1 - (0.841)^2}} = 3.108$$

$$\therefore |t| < t_{n-2}$$

$$\Rightarrow |3.108| < t_{6-2}$$

$$\Rightarrow |3.108| < t_4$$

$$\Rightarrow |3.108| < 2.776$$

$\therefore H_0$ is Rejected

So, Lending rate increase significantly with the increase of inflation rate.

③

We know—

$$\hat{y} = a + bx \quad \text{--- (i)}$$

$$a = \bar{y} - b\bar{x} \quad \text{--- (ii)}$$

$$b = \frac{SP(xy)}{SS(x)}$$

$$= \frac{\sum xy - \frac{\sum x \sum y}{n}}{SS(x)} = \frac{2036.68 - \frac{104.4 \times 158}{6}}{116.56}$$

$$= 1.6084$$

put the values in eq (ii) —

$$a = \frac{158}{6} - 1.6084 \times \frac{104.4}{6}$$

$$= -1.652$$

$$[\because \bar{y} = \frac{\sum y}{n}; \bar{x} = \frac{\sum x}{n}]$$

Now, put the values in eq (1) —

$$\hat{y} = a + bx$$

$$= -1.652 + 1.6084x$$

①

In given —

$$x = 25.5$$

we know —

$$\hat{y} = a + bx$$

$$= -1.652 + 1.608 \times 25.5$$

$$= 39.352$$

②

⇒ we need to test —

$$H_0: \beta = 0 \quad \text{Vs} \quad H_1: \beta \neq 0$$

we know —

$$t = \frac{b}{\sqrt{S^2/SS(x)}} \quad \text{--- (1)}$$

$$S^2 = \frac{SS(y) - bSS(x)}{n-2}$$

$$= \frac{425.09 - 1.6084 \times 116.56}{6-2}$$

$$= 59.62$$

put it into eq (1) —

$$t = \frac{1.6084}{\sqrt{59.62/116.56}}$$

$$= 2.248$$

$$\therefore |t| < t_{n-2}$$

$$|2.248| < t_{6-2}$$

$$|2.248| < 2.776$$

∴ H_0 is accepted

So the regression isn't significant